



VIT[®]

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

FALL SEMESTER 2025-26

CSI3007 - Advanced Python Programming

SLOT: A2

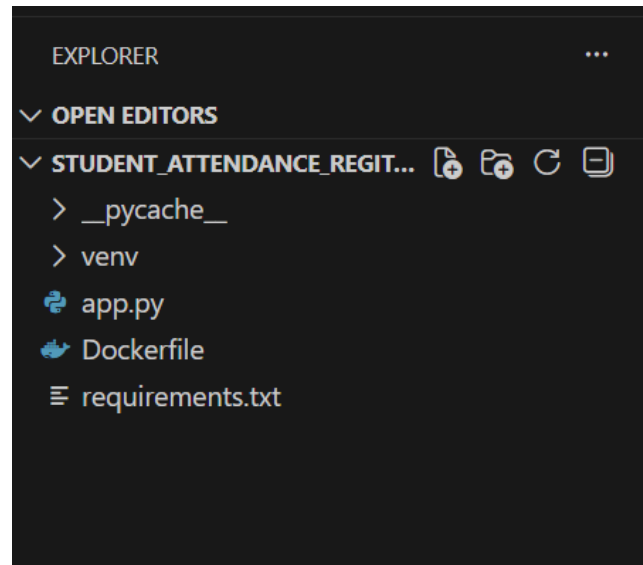
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REG NO.: 22MID0300

Date: 14-11-2025

Student Attendance Management API using FastAPI and Cloud Deployment

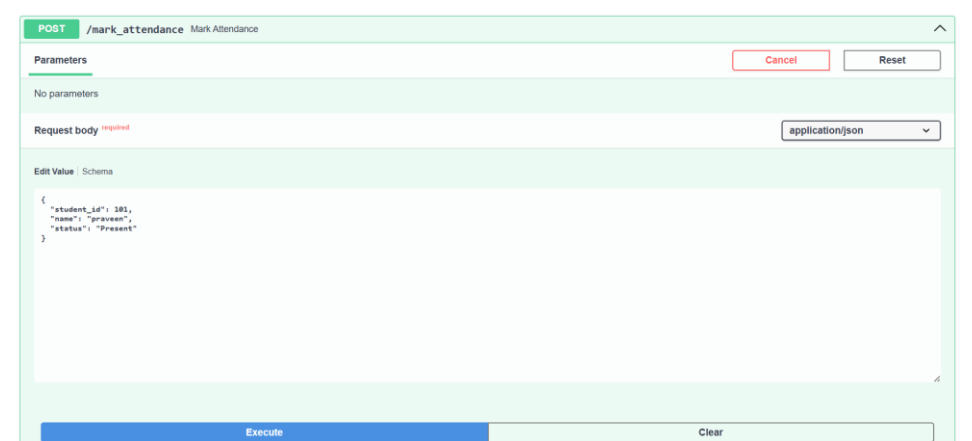
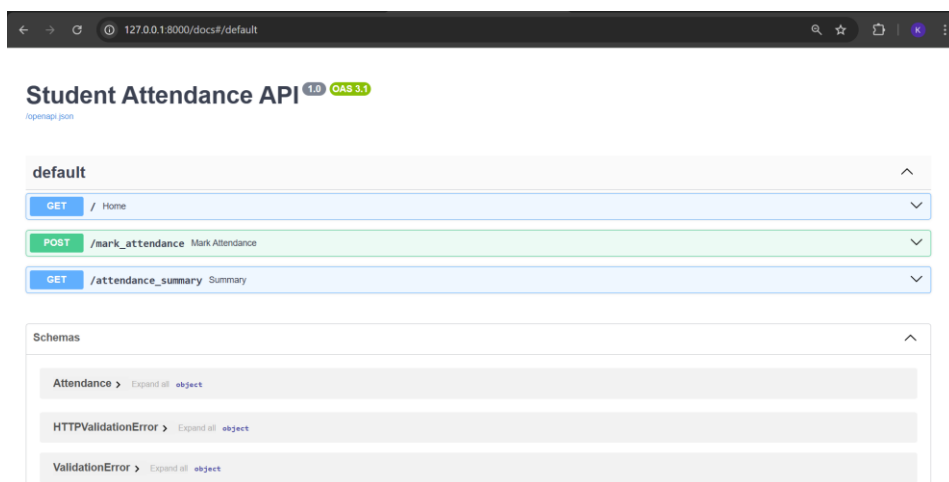
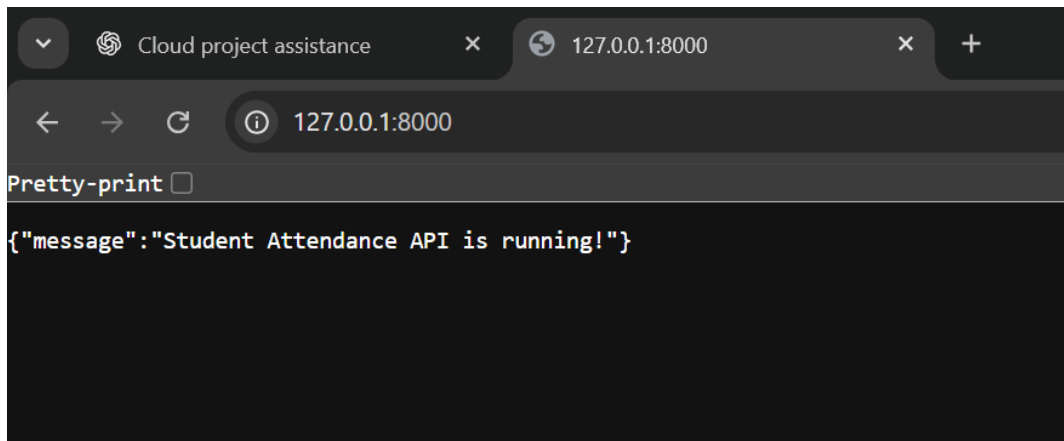


```
app.py 4 X
app.py > ...
1  from fastapi import FastAPI, HTTPException, Request
2  from fastapi.middleware.cors import CORSMiddleware
3  from pydantic import BaseModel
4  from mangum import Mangum
5
6  # Create API
7  app = FastAPI(title="Student Attendance API", version="1.0")
8
9  # Enable CORS
10 app.add_middleware(
11     CORSMiddleware,
12     allow_origins=["*"],
13     allow_credentials=True,
14     allow_methods=["*"],
15     allow_headers=["*"],
16 )
17
18 # -----
19 # Request Model
```

```
Dockerfile X
Dockerfile
1 FROM public.ecr.aws/lambda/python:3.11
2
3 WORKDIR /var/task
4
5 RUN yum install -y gcc make rust cargo && \
6     python3.11 -m pip install --upgrade pip setuptools wheel
7
8 COPY requirements.txt .
9 RUN python3.11 -m pip install --no-cache-dir -r requirements.txt
10
11 COPY . .
12
13 CMD ["app.lambda_handler"]
14
```

```
requirements.txt X
requirements.txt
1 fastapi
2 uvicorn
3 mangum
4 python-multipart
5
```

```
(venv) PS C:\Users\hp\Desktop\python\student_attendance_regitar_api> uvicorn app:app --reload
>>
INFO: Will watch for changes in these directories: ['C:\\Users\\hp\\Desktop\\python\\student_
e_regitar_api']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [3400] using StatReload
INFO: Started server process [12716]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:41080 - "GET / HTTP/1.1" 200 OK
```



Responses

Curl

```
curl -X 'POST' \  
  'http://127.0.0.1:8000/mark_attendance' \  
  -H 'accept: application/json' \  
  -H 'Content-Type: application/json' \  
  -d '{  
    "student_id": 101,  
    "name": "praveen",  
    "status": "Present"  
  }'  
,
```

Request URL

```
http://127.0.0.1:8000/mark_attendance
```